JOM, Vol. 75, No. 4, 2023
https://doi.org/10.1007/s11837-022-05650-0

TECHNICAL ARTICLE

Magnetic Property Improvement and Core-Loss Reduction of Fe-Si-Cr-Based Soft Magnetic Composites with Addition of Fe-50Ni Nanopowder

YEON JUN CHOI,[1] MIN YOUNG LEE,[1] and BO WHA LEE [1,2]

1.—Department of Physics and Oxide Research Center, Hankuk University of Foreign Studies, Yongin 17035, South Korea. 2.—e-mail: bwlee@hufs.ac.kr

We studied the effect of change in packing fraction, caused by the addition of nanopowder to micro-sized powders, on the magnetic permeability and core loss of a power inductor core in the high-frequency band. Fe-Si-Cr crystalline alloy was used as the base material of the power inductor core, and Fe-50Ni nanopowder produced by the pulsed wire evaporation method was added to the alloy powder. With the addition of only 10 wt.% Fe-50Ni nanopowder, the core showed the maximum packing fraction and good power inductor characteristics, along with the maximum permeability of 26.598 H/m. Furthermore, the permeability and packing fraction showed the same tendency. In addition, the loss of the core containing 10 wt.% of Fe-50Ni nanopowder was 210.88 mW/cm^3, which is $\sim$ 78.91% lower than that of pure Fe-Si-Cr core. The results obtained in this study reveal that the characteristics of a power inductor can be significantly modified by changing the packing fraction of the core.

INTRODUCTION

Currently, various types of electronic components are used in electronic devices. Among them, power inductors are key components that supply power to electronic devices. Efforts have been made to improve the characteristics of power inductors, such that a higher stability and an improved permeability in the 1-MHz high-frequency band as well as low core losses are realized.[1–4] To achieve such advantageous characteristics, many soft magnetic materials have been studied, and recently soft magnetic composites (SMCs) mixed with insulating polymers have been used as a good power inductor material.[5,6]

In general, Fe-based metal powder with a high saturation magnetization exhibits good DC bias characteristics, because it maintains inductance at a high current and has a high current capacity.[7,8] In addition, according to Snoek's limitation, magnetic resonance frequency appears in the megahertz band; therefore, Fe-based metal powder is widely used for applications in the high-frequency band.[7,8] However, the eddy current losses also increase because of its lower resistivity compared with those of ferrite materials, which have been widely used in the past.[9–11] Furthermore, the increase in eddy current losses due to the I^2R losses generates heat in the magnetic material, which adversely affects electronic products, such as inductors and transformers.[12]

To overcome the disadvantages of the Fe-based metal powder, studies on amorphous materials with high resistivity and good power inductor characteristics have been conducted.[13–17] However, amorphous materials have a lower saturation flux density than Si steel, which is an alloy of Fe and Si.[18] In addition, amorphous materials are difficult and expensive to manufacture. On the other hand, Fe-based crystalline alloys are easy to manufacture and supply, and also exhibit a higher saturation magnetization value than the amorphous alloy.[19–22] In our previous study on Fe-Si-Cr + carbonyl iron powder (CIP) SMCs, we confirmed the effect of reducing the core loss of crystalline alloys.[23] In this study, we analyzed the effect of adding nanopowders to Fe-Si-Cr powder.

(Received March 22, 2022; Revised /revised November 8, 2022; accepted December 9, 2022;
published online January 17, 2023)

To overcome the disadvantage of all-metallic powders having relatively low magnetic permeability compared to metal oxide powders, such as ferrites, we employed a method of improving the magnetic permeability by increasing the packing fraction based on Ollendorff's equation.[24] To improve the packing fraction, we used a mixing powder, Fe-50Ni nanopowder, prepared by pulsed wire evaporation (PWE) with relatively smaller sized particles than those of the Fe-Si-Cr powders. The nanopowders produced using the PWE method contain particles with a uniform diameter, and an oxidation layer, which increases the resistivity, is formed on the surface of the powder.[25,26] Furthermore, the PWE method is highly energy efficient, environment friendly, and requires a simple apparatus for powder fabrication.[26,27] The schematic of the PWE method is displayed in Fig. 1. The wire is evaporated instantaneously by a pulsed electric current, creating plasma conditions. The plasma then interacts with a coolant gas in the chamber, and the nanopowders are condensed.[28–30] In addition, the Fe-50Ni material containing ≥ 50 wt.% Ni exhibits a high permeability and low coercive force and is thus expected to improve the power inductor characteristics. Thus, the Fe-50Ni nanopowder produced by PWE was selected as the additional material in this study.[26]

The samples were prepared by mixing the PWE-synthesized Fe-50Ni nanopowders with Fe-Si-Cr powder in various ratios, and the change in the permeability according to the improvement in the packing fraction was investigated. Additionally, the changes in the total core loss according to the variations in the eddy current loss were evaluated.

EXPERIMENTAL DETAILS

Powder Analysis

The Fe-Si-Cr powder was prepared by gas atomization and was subsequently sieved to obtain a powder with particles < 25 μm in size. The Fe-50Ni nanopowders were prepared using the PWE method, and the intrinsic properties of each powder were analyzed. Scanning electron microscopy (SEM; JEOL JEM-7600F) was used to analyze the morphology of the powders, and energy dispersive x-ray spectroscopy (EDS; Oxford Instruments JSM-6390) was performed to identify the atomic constitution of the fabricated powders. For SEM measurement, 5 kV was applied for the beam voltage. For EDS analysis, we measured the spectra of the powder and analyzed the atomic constitution of each powder in terms of wt.% value. X-ray diffraction (XRD; Rigaku Miniflex x-ray diffractometer) was performed to determine the crystal structures of the powders, and a vibrating sample magnetometer (VSM; Lakeshore VSM7404 series) was used to analyze their magnetic properties.

Fabrication of Toroidal Cores

To measure the magnetic permeability and core loss of the samples in the high-frequency band, toroidal cores were made with different ratios of Fe-50Ni nanopowders. In this study, 10 wt.%, 20 wt.%, 30 wt.%, and 40 wt.% Fe-50Ni nanopowders were mixed with the Fe-Si-Cr powder, which was further mixed with 3.0 wt.% of an insulating epoxy resin. The sample containing all the mixed materials was placed in a toroidal core-type mold and pressed under a pressure of 3 tons for 1 min. The outer and inner diameters of the toroidal core were 20 mm and 13 mm, respectively, and the pressed core was heat treated at 200 °C for 2 h.

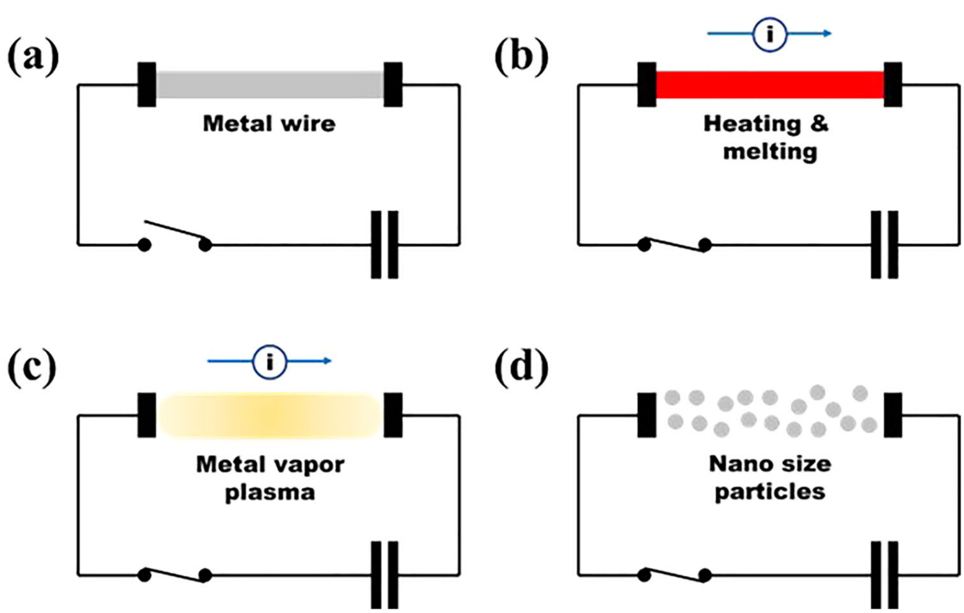

Fig. 1. Schematic of the PWE method.

Measurement of the Toroidal Core Properties

To measure the permeability and quality factor (Q-factor) of the manufactured core, a Cu wire was wound around the core ten times to prepare a sample. An impedance analyzer (Hewlett Packard, 4294A) was used to measure the core permeability and Q-factor from 100 kHz to 110 MHz. A B–H analyzer (Iwatsu, SY-8219) was used to measure the core loss from 100 kHz to 1 MHz.

RESULTS AND DISCUSSION

We observed the morphology of the Fe-Si-Cr powder and Fe-50Ni nanopowder using SEM as shown in Fig. 2. Figure 2a and b shows the morphologies of the PWE-synthesized Fe-50Ni nanopowder and Fe-Si-Cr powder, respectively. The average size of the Fe-50Ni nanopowder particles was ~ 70 nm, and the Fe-Si-Cr powder was sieved to retain particles with sizes < 25 μm. The particle size distribution of the Fe-Si-Cr powder is shown in Fig. 2c. Both the Fe-50Ni nanopowder and Fe-Si-Cr powder contained spherical particles. In particular, it can be inferred that the spherical particles of the Fe-Si-Cr powder originated from its manufacturing process, i.e., gas atomization. According to the particle packing theory, a cubic body has a maximum density when spherical particles are perfectly layered in a body-centered cubic with a maximum packing fraction of approximately 74%.[31] Suzuki et al.[32] proposed that the packing fraction of the particle-accumulated bulk body can be defined as the total volume excluding porosity. Therefore, based on the results of the previous studies, we expected that the packing fraction could be improved by adding nanopowders to micro-sized powders.

Table I shows the atomic constitution of each powder, measured by EDS. In the Fe-50Ni nanopowder, ~ 45.54 wt.% of Fe and ~ 50.09 wt.%

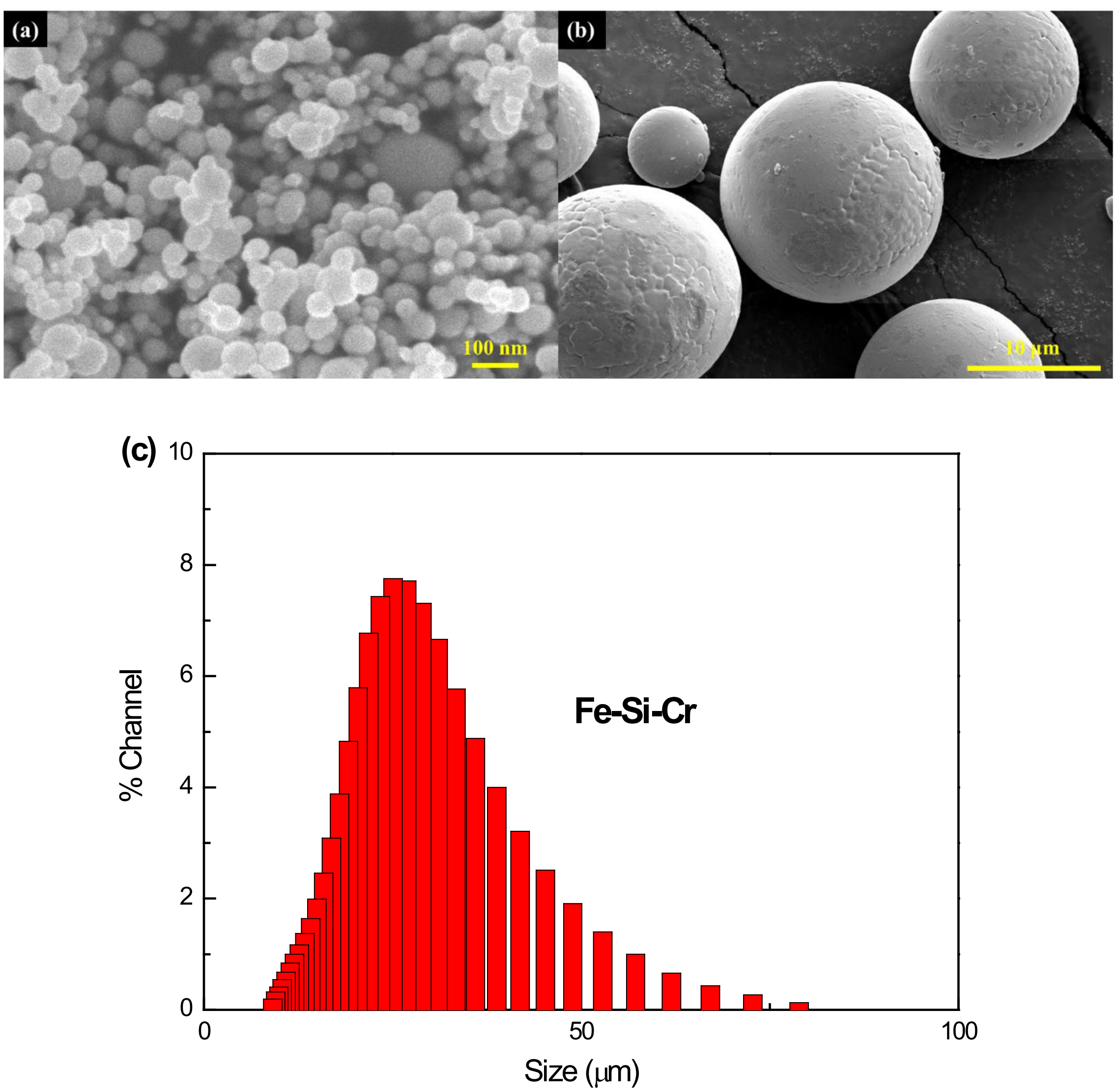

Fig. 2. Morphologies of the (a) Fe-50Ni nanopowder and (b) Fe-Si-Cr powder. (c) Particle size distribution of Fe-Si-Cr powder.

of Ni were detected, whereas in the Fe-Si-Cr powder, 87.28 wt.%, 10.13 wt.%, and 1.94 wt.% of Fe, Si, and Cr were detected, respectively. The standard variations of Fe-50Ni nanopowder were 1.01, 0.56, and 0.19 for Fe, Ni, and O, respectively. The standard variations of Fe-Si-Cr were 1.01, 0.88, 0.30, and 0.20 for Fe, Si, Cr, and O, respectively. Oxygen was detected in both powders, approximately 4.38 wt.% in the Fe-50Ni nanopowder and 0.65 wt.% in the Fe-Si-Cr powder, owing to the oxidation layer that appeared on the powder surface during the PWE process. In the case of the Fe-Si-Cr powder, a Cr-rich oxidation layer was formed because of the addition of Cr.[33–35] The EDS spectra of the Fe-50Ni nanopowder and Fe-Si-Cr are shown in Fig. 3.

Table I. EDS results of the Fe-50Ni nanopowder and Fe-Si-Cr powder

Contents (wt.%)	Fe	Ni	Si	Cr	O
Fe-50Ni	45.53	50.09	–	–	4.38
Fe-Si-Cr	87.28	–	10.13	1.94	0.65

Figure 4 presents the XRD patterns indicating the crystal structure of each powder. The Fe-50Ni nanopowder had a face-centered cubic structure and exhibited XRD peaks corresponding to the (111), (200), and (220) directions, indicating a *Fm3m* space group symmetry, which is consistent with JCPDS #47-1417 data.[36] In contrast, the XRD spectrum of the Fe-Si-Cr powder showed peaks corresponding to the (110), (200), and (211) directions, indicating a body-centered cubic structure. The corresponding crystal structure was represented by a *Im3m* space group, consistent with the JCPDS #06-0696 data: an α-Fe XRD peak was observed as well.[37]

Figure 5 shows the magnetic hysteresis loops of the Fe-50Ni nanopowder and Fe-Si-Cr powder, measured using a VSM. Both powders showed good soft magnetic properties and had low coercivity (H_c) and saturation magnetization (M_s). H_c and M_s of the Fe-50Ni nanopowder were 323.13 Oe and 121.18 emu/g, respectively, whereas those of the Fe-Si-Cr powder were 6.92 Oe and 157.07 emu/g,

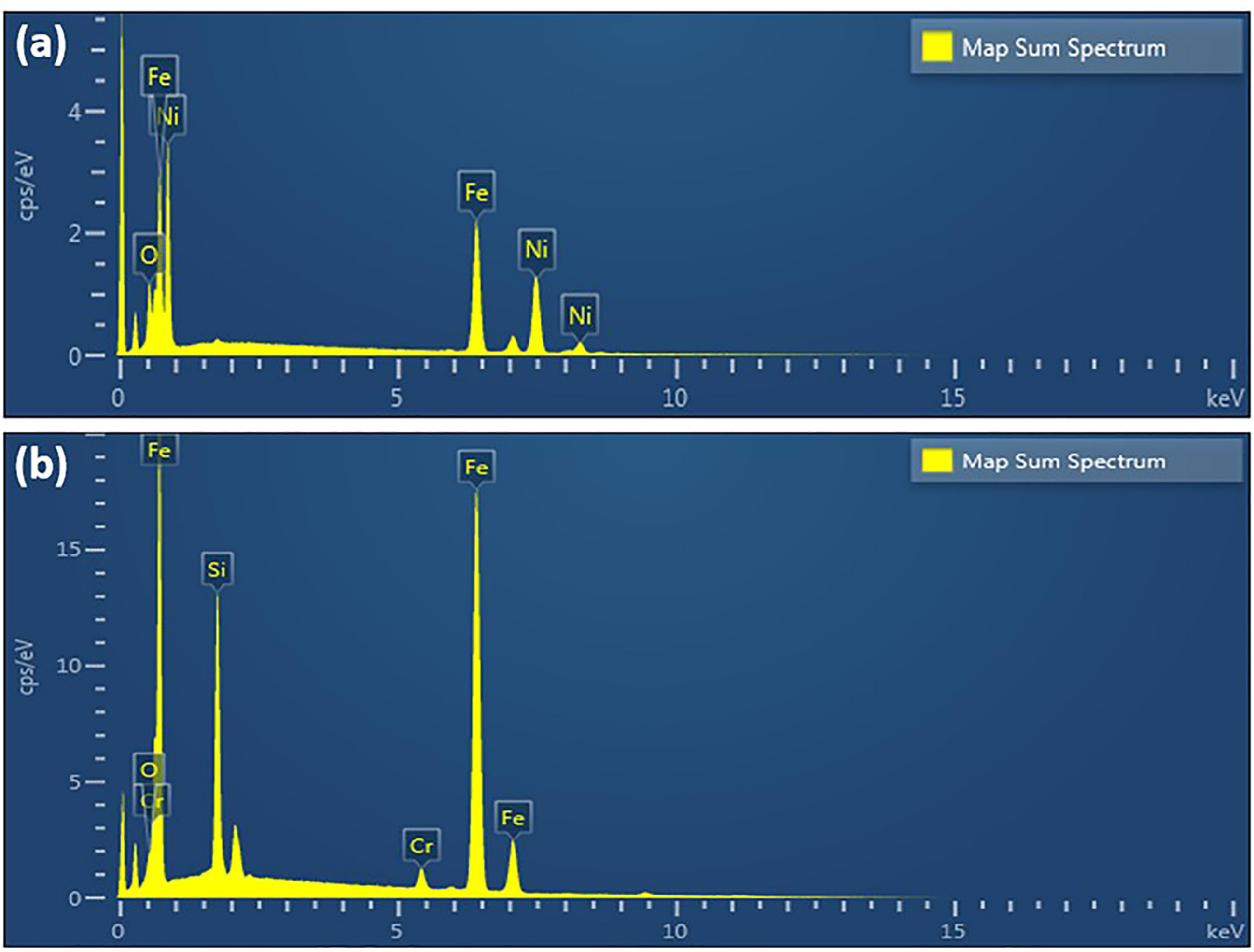

Fig. 3. EDS spectra of (a) the Fe-50Ni nanopowder and (b) the Fe-Si-Cr powder.

respectively. The M_s of the Fe-50Ni nanopowder was lower than that of the Fe-Si-Cr powder because the content of Fe, which has a high M_s value, in the Fe-50Ni nanopowder was lower than that in the Fe-Si-Cr powder. In addition, the H_c of the Fe-50Ni nanopowder was higher than that of the Fe-Si-Cr powder because as the particle size decreases to 76 nm or less, the powder acts as a superparamagnetic material in a low-strength magnetic field region, i.e., the powder acts as a single domain rather than a multidomain.[38]

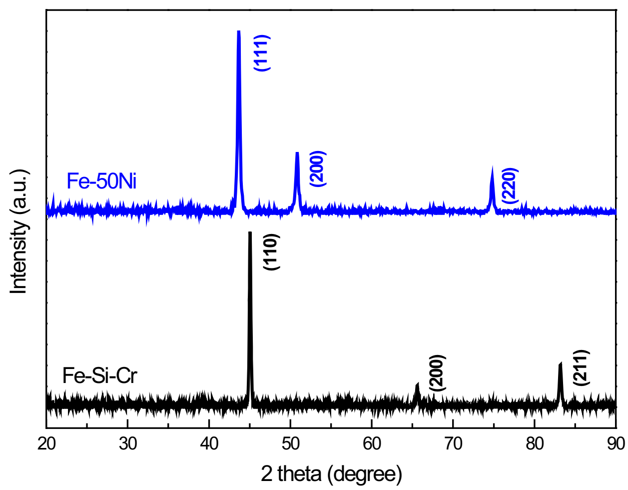

Fig. 4. XRD patterns of the Fe-50Ni nanopowder and Fe-Si-Cr powder.

Figure 6 shows the permeability and Q-factor of the composites in the 100 kHz to 110 MHz band, measured with an impedance analyzer. The real part of the permeability, shown in Fig. 6a, indicates that the magnetic permeability of the Fe-Si-Cr powder rapidly increased with an addition of only 10 wt.% of Fe-50Ni nanopowder. Most of the composites showed a higher permeability than the core, which was made of pure Fe-Si-Cr powder. However, in the case of the composite containing 40 wt.% of Fe-50Ni nanopowder, the magnetic permeability slightly decreased. Figure 6b shows the imaginary part of the permeability, revealing that the imaginary part of the permeability of the composites containing 30 wt.% and 40 wt.% of Fe-50Ni nanopowder was higher than that of the other samples. Figure 6c shows the Q-factor value, which is an index used to evaluate the efficiency of a power inductor, defined as:

$$Q = \frac{\mu'}{\mu''} = \frac{1}{\tan\delta} \tag{1}$$

where μ' and μ'' represent the permeabilities, and tan $\tan\delta$ is the loss tangent. The Q-factor is directly related to the permeability of the core and is the reciprocal of the loss tangent.[38] The samples containing 30 wt.% and 40 wt.% of Fe-50Ni nanopowder showed a decrease in the Q-factor value, while their resonant frequency increased compared to that of the other samples, because of the relatively high imaginary-part permeability of these samples

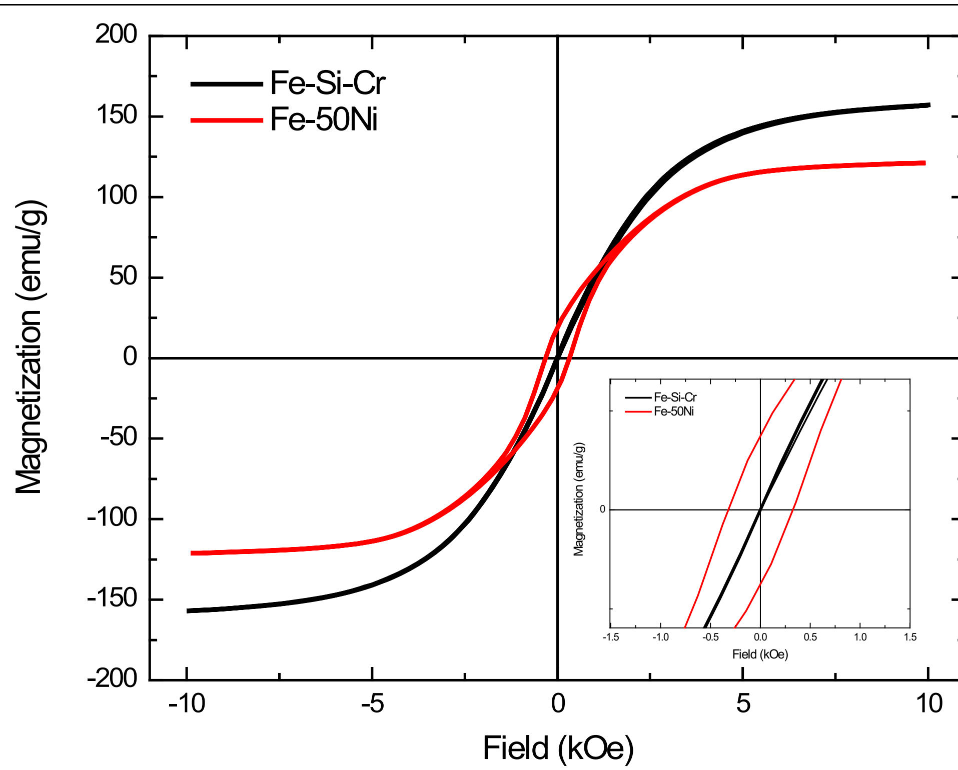

Fig. 5. Magnetic hysteresis loops of the Fe-50Ni nanopowder and Fe-Si-Cr powder. (Inset) Zoomed-in MH curve of the powders.

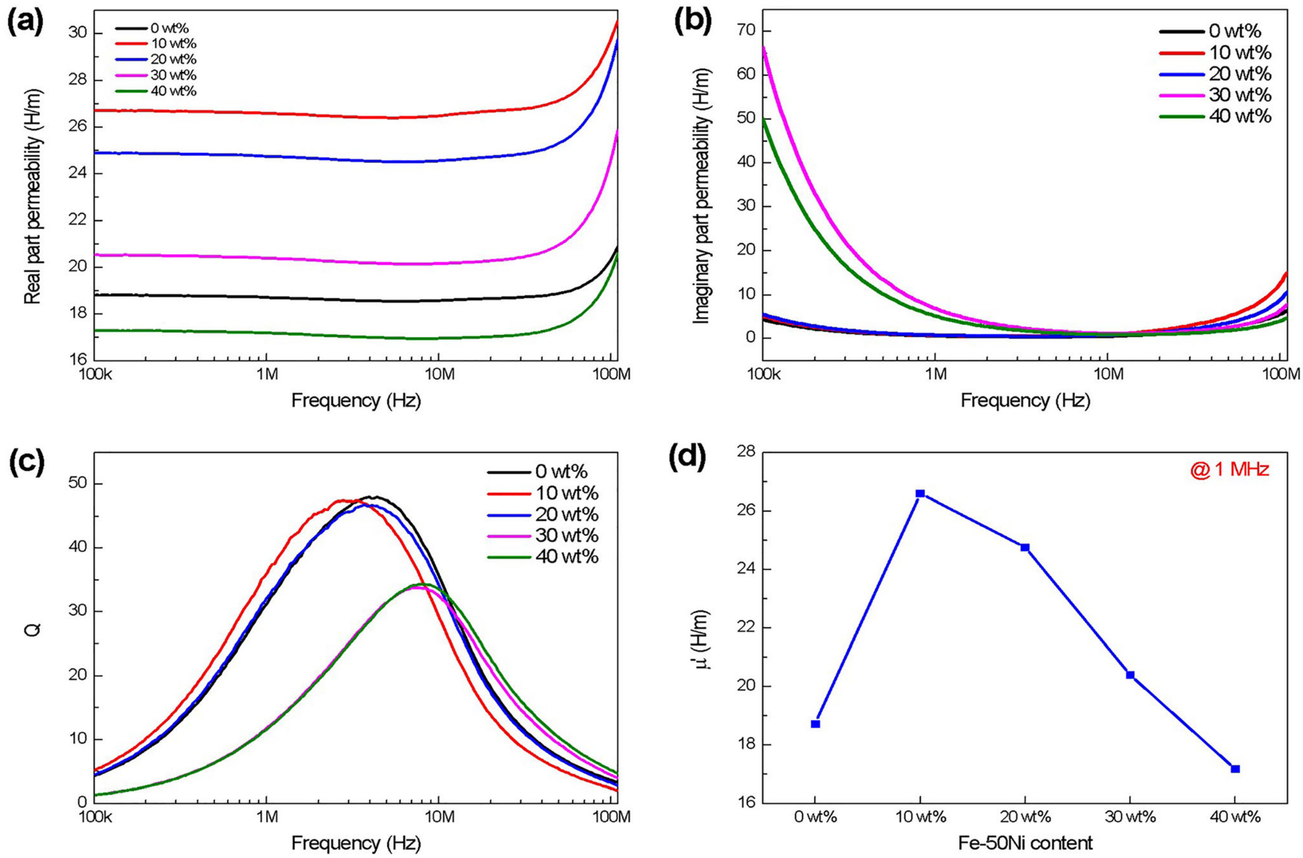

Fig. 6. (a) Real-part and (b) imaginary-part permeability. (c) Q-factor as a function of the applied frequency from 100 kHz to 110 MHz. (d) Real-part permeability at 1 MHz.

in the low-frequency band. Figure 6d shows the real-part permeability of each sample at 1 MHz. The sample containing 10 wt.% Fe-50Ni nanopowder showed a sharply increased magnetic permeability from 18.715 H/m to 26.598 H/m compared to the sample without nanopowder.

Figure 7 indicates the relationship between core density and permeability. The core with the highest density showed the highest permeability, and it was found that the core density and permeability exhibited the same tendency. In general, the density increased with the addition of the Fe-50Ni nanopowder; in turn, the magnetic permeability increased as well. However, the sample containing 40 wt.% of Fe-50Ni nanopowder showed a reduced density, resulting in a corresponding decrease in the magnetic permeability. As the density increases, the air gap decreases, and the permeability is improved. Thus, a reduced core loss can be expected.[23,39,40]

Table II shows the core density, real-part permeability, imaginary-part permeability, and Q-factor values at 1 MHz for each sample. It could be verified that a higher core density results in a higher real-part permeability and Q-factor.

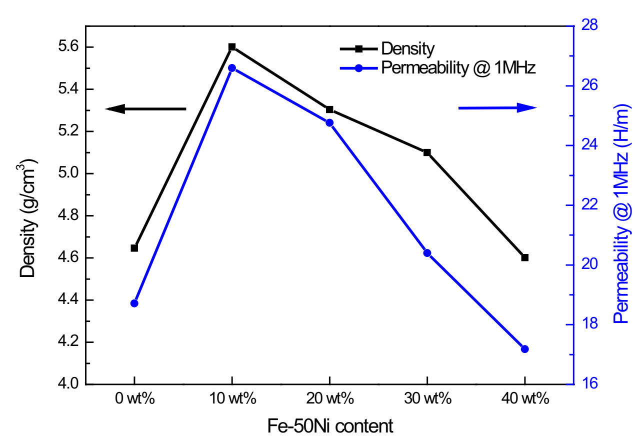

Fig. 7. Relationship between the density of each core and permeability at 1 MHz.

Figure 8 shows the relationship between the core density and packing fraction of the composites, and a comparison with the theoretical results obtained using Ollendorff's equation. Figure 8a shows a graph comparing the density and packing fraction of the cores. Equation 2 was used to calculate the packing fraction:[41]

$$\text{Packing faction} = \left[\frac{d_{\text{core}}}{d_{\text{bulk}}} \times 100\right]\% \quad (2)$$

where d_{core} and d_{bulk} are the core and bulk densities, respectively. The calculated relationship between packing fraction and density showed the same trend. Furthermore, the sample containing 10 wt.% Fe-50Ni nanopowder showed the highest value with a packing fraction of 77.24%. Figure 8b shows a comparison between the experimental data and the values calculated using Ollendorff's equation, which expresses the relationship between packing fraction and permeability as shown in Eq. 3. Notably, high permeability is obtained at a high packing fraction.[24]

$$\mu = \mu_0 + \frac{\eta\mu_0(\mu_m - \mu_0)}{N(1-\eta)(\mu_m - \mu_0) + \mu_0} \quad (3)$$

Table II. Characteristics of the Fe-50Ni-nanopowder-added Fe-Si-Cr powder-based cores, measured using an impedance analyzer

@ 1 MHz	ρ_b (g/cm^3)	μ' (H/m)	μ'' (H/m)	Q
0 wt.%	4.646	18.715	0.597	31.362
10 wt.%	5.601	26.598	0.737	36.110
20 wt.%	5.304	24.758	0.769	32.213
30 wt.%	5.100	20.399	6.801	11.848
40 wt.%	4.601	17.183	5.124	11.676

The above equation shows that the permeability of the core is affected by three main factors: η (volume packing fraction), N (coefficient of demagnetization factor, $0 \le N \le 1$), and μ_m (permeability of powder). μ_0 is the magnetic permeability in vacuum and is given by $\mu_0 = 4\pi \times 10^{-7}$ H/m. In our study, $N = \frac{1}{3}\exp(-3\eta)$ was used for the calculation, based on the report of Anhalt et al.[42] The values calculated using Ollendorff's equation were compared to the experimentally obtained result of the pure Fe-Si-Cr core. Although the calculated values and experimental data showed the same tendency, they did not show a complete matching, and the magnetic permeability increased more rapidly with increasing packing fraction. It seems that the inhomogeneity in particle size occurs because of the addition of the Fe-50Ni nanopowder, and the experimentally analyzed relationship between magnetic permeability and packing fraction, showing a relatively steeper increase, is significantly deviated from the calculated one, obtained using Ollendorff's equation.[43]

Figure 9 shows the core loss from 100 kHz to 1 MHz, measured using a B–H analyzer at B_m = 10 mT. The total core loss is composed of a hysteresis loss (P_{h}), an eddy current loss (P_{e}), and a residual loss (P_{r}) and can be defined by Eq. 4:

$$P_{\text{cv}} = P_{\text{h}} + P_{\text{e}} + P_{\text{r}} \cong P_{\text{h}} + P_{\text{e}} \cong W_{\text{h}}f + W_{\text{e}}f \quad (4)$$

where W_{h} and W_{e} are the hysteresis and eddy current loss constants, respectively. In the case of a metallic magnetic core, the residual loss is exceptionally low and therefore negligible.[44] The core loss increases linearly with increasing frequency. In Fig. 9a, it can be seen that the loss of the core, which contained the Fe-50Ni nanopowder, occurred more rapidly than that of the pure Fe-Si-Cr. In particular, the sample containing 10 wt.% Fe-50Ni

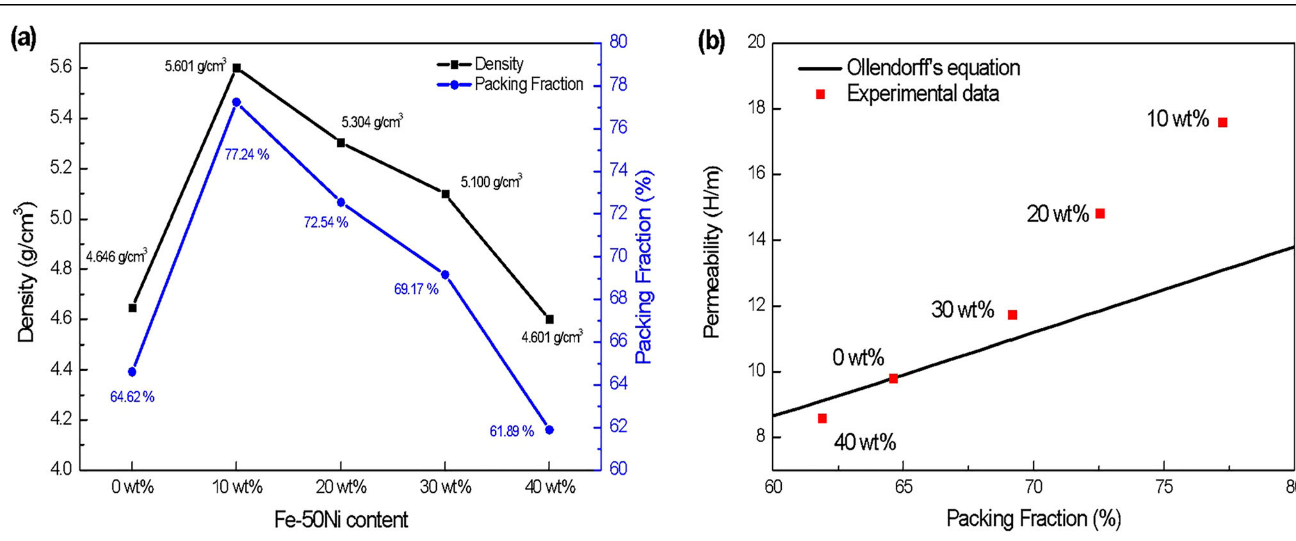

Fig. 8. (a) Relationship between the density and packing fraction of the cores and (b) comparison between calculated and experimentally obtained permeability as a function of the packing fraction.

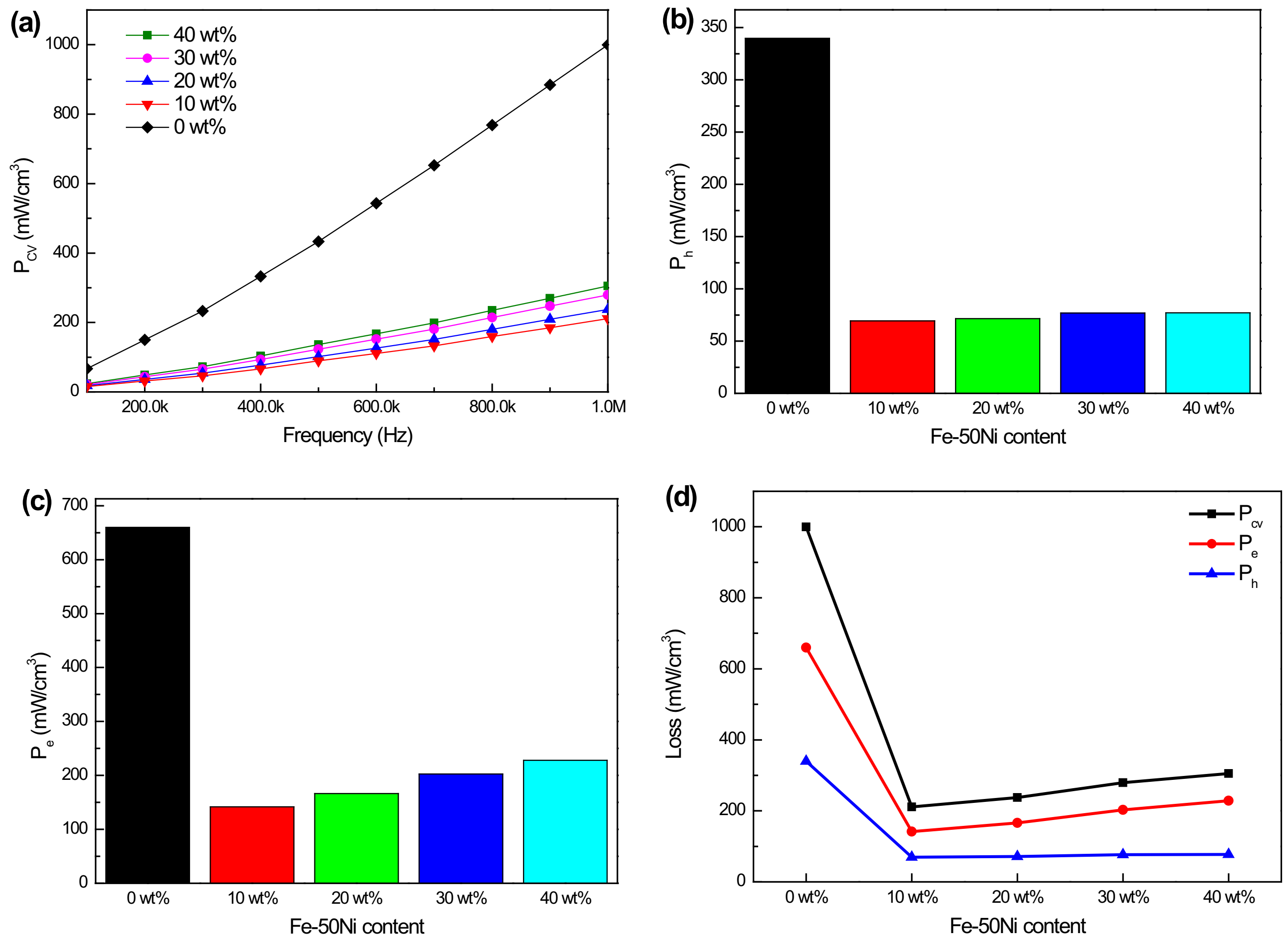

Fig. 9. (a) Total core loss, (b) hysteresis loss, and (c) eddy current loss of each sample, measured from 100 kHz to 1 MHz and (d) those measured at 1 MHz for each sample.

nanopowder, which showed the highest packing fraction and permeability, exhibited the lowest core loss; only 10 wt.% of the Fe-50Ni nanopowder was added, and the core loss drastically decreased from 999.82 mW/cm^3 to 210.88 mW/cm^3 at 1 MHz. It is inferred that the addition of the Fe-50Ni nanopowder reduced the core loss, because the addition of other powders reduced the intraparticle loss. This resulted in an increase in the overall resistivity of the core, which reduced the eddy current loss, ultimately leading to a decrease in the overall core loss. Figure 9b and c shows the hysteresis loss and eddy current loss of the core, respectively. Separation of hysteresis loss and eddy current loss from the total core loss was performed through the curve fitting method. For curve fitting, we used OriginLab software and conducted linear fitting on the core loss data. The hysteresis loss of the samples containing 10–40 wt.% of Fe-50Ni nanopowder showed a negligible difference, whereas the eddy current loss was affected by the amount of added Fe-50Ni nanopowder. This indicates that the effect of the eddy current loss is greater than that of the hysteresis loss on the reduction of the total core loss. In Fig. 9d, the core loss, hysteresis loss, and eddy current loss at 1 MHz are compared. Thus, it was confirmed that the sample containing 10 wt.% of Fe-50Ni nanopowder showed the highest packing fraction and permeability and exhibited the lowest core loss and eddy current loss.

CONCLUSION

In this study, we investigated the changes in the AC magnetic properties of Fe-Si-Cr powder-based SMCs in high-frequency bands with the addition of Fe-50Ni nanopowder. The sample containing 10 wt.% of Fe-50Ni nanopowder exhibited the highest packing fraction of 77.24% and magnetic permeability of 26.598 H/m at 1 MHz, showing the same trend as that obtained using Ollendorff's equation. In addition, this sample showed the lowest core loss of 210.88 mW/cm^3 at 1 MHz, which is equivalent to a reduction of $\sim$ 78.91% compared to that shown by the pure Fe-Si-Cr core. These experimental results reveal that the power inductor characteristics of the core can be changed by changing the intrinsic properties of the powders constituting the SMC as well as the microstructure represented by the packing fraction. Additionally, a core showing a high packing fraction exhibits a high magnetic permeability and low core loss.

ACKNOWLEDGEMENTS

This research was supported by Materials, Components & Equipment Research Program funded by the Gyeonggi Province (AICT-003-T1).

CONFLICT OF INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

REFERENCES

1. A. Goldman, in *Handbook of Modern Ferromagnetic Materials* (Springer US, Boston, MA, 1999), pp. 183–205.
2. J.M. Silveyra, E. Ferrara, D.L. Huber, and T.C. Monson, *Science (1979)* 362, 418 (2018).
3. L.O. Hultman and A. G. Jack, in *IEEE International Electric Machines and Drives Conference, 2003. IEMDC'03.* (IEEE, n.d.), pp. 516–522.
4. I.P. Gilbert, V. Moorthy, S.J. Bull, J.T. Evans, and A.G. Jack, *J. Magn. Magn. Mater.* 242–245, 232 (2002).
5. H. Shokrollahi and K. Janghorban, *J. Mater. Process Technol.* 189, 1 (2007).
6. A. Talaat, M.V. Suraj, K. Byerly, A. Wang, Y. Wang, J.K. Lee, and P.R. Ohodnicki, *J. Alloys Compd.* 870, 159500 (2021).
7. T. Ueno, H. Tsuruta, T. Saito, A. Watanabe, T. Ishimine, and K. Yamada, *SEI Tech. Rev.* 82, 9 (2016).
8. J.L. Snoek, *Physica* 14, 207 (1948).
9. Kh. Gheisari, S. Javadpour, H. Shokrollahi, and B. Hashemi, *J. Magn. Magn. Mater.* 320, 1544 (2008).
10. A.K.M.A. Hossain, S.T. Mahmud, M. Seki, T. Kawai, and H. Tabata, *J. Magn. Magn. Mater.* 312, 210 (2007).
11. J.H. Paterson, R. Devine, and A.D.R. Phelps, *J. Magn. Magn. Mater.* 196–197, 394 (1999).
12. A.M. Leary, P.R. Ohodnicki, and M.E. McHenry, *JOM* 64, 772 (2012).
13. T. Zhao, C. Chen, X. Wu, C. Zhang, A.A. Volinsky, and J. Hao, *J. Alloys Compd.* 857, 157991 (2021).
14. N. Yabu, K. Sugimura, M. Sonehara, and T. Sato, *IEEE Trans. Magn.* 54, 1 (2018).
15. D. Azuma, N. Ito, and M. Ohta, *J. Magn. Magn. Mater.* 501, 166373 (2020).
16. C. Chang, Y. Dong, M. Liu, H. Guo, Q. Xiao, and Y. Zhang, *J. Alloys Compd.* 766, 959 (2018).
17. D. Azuma and R. Hasegawa, *IEEE Trans. Magn.* 47, 3460 (2011).
18. H.R. Lashgari, D. Chu, S. Xie, H. Sun, M. Ferry, and S. Li, *J. Non Cryst. Solids* 391, 61 (2014).
19. A. Ahmad, S. Mitra, S.K. Srivastava, and A.K. Das, *J. Magn. Magn. Mater.* 540, 168449 (2021).
20. A. Ahmad, S.K. Srivastava, and A.K. Das, *J. Alloys Compd.* 878, 160341 (2021).
21. S. Mitra, A. Ahmad, S. Chakrabarti, S. Biswas, and A.K. Das, *J. Magn. Magn. Mater.* 552, 169148 (2022).
22. A. Ahmad, S. Mitra, S.K. Srivastava, and A.K. Das, *J. Magn. Magn. Mater.* 474, 599 (2019).
23. Y.J. Choi, J.H. Ahn, S.W. Kim, Y.R. Kim, and B.W. Lee, *MRS Commun.* 11, 457 (2021).
24. F. Ollendorff, *Archiv Für Elektrotechnik* 25, 436 (1931).
25. H. Suematsu, Y. Hayashi, N.D. Hieu, T.M.D. Do, and T. Nakayama, *Jpn. J. Appl. Phys.* 40, 1100 (2021).
26. W. Jiang and K. Yatsui, *IEEE Trans. Plasma Sci.* 26, 1498 (1998).
27. Y. Kinemuchi, T. Ikeuchi, T. Suzuki, H. Suematsu, W. Jiang, and K. Yatsui, *IEEE Trans. Plasma Sci.* 30, 1858 (2002).
28. H.-S. Seo and H.S. Park, *J. Nanosci. Nanotechnol.* 18, 6823 (2018).
29. D.A.H. Hanaor and C.C. Sorrell, *J. Mater. Sci.* 46, 855 (2011).
30. W. Wunderlich, *Electronic Properties of Nano-Porous TiO2- and ZnO-Thin Films-Comparison of Simulations and Experiments Estimation and Calculation of Electron-Phonon-Interaction (EPI) Parameter for Alloys View Project Properties of Organic Semiconductor Thin Films View Project* (2004).
31. S. Li, J. Zhao, P. Lu, and Y. Xie, *Chin. Sci. Bull.* 55, 114 (2010).
32. M. Suzuki, H. Kada, and M. Hirota, *Adv. Powder Technol.* 10, 353 (1999).
33. H.-I. Hsiang, *J. Mater. Sci.: Mater. Electron.* 31, 16089 (2020).
34. H. Matsuura and K. Otake, US8416051B2 (2013).
35. T.D. Zhou, J.K. Tang, and Z.Y. Wang, *J. Magn. Magn. Mater.* 322, 2589 (2010).
36. Y. Feng and T. Qiu, *J. Magn. Magn. Mater.* 324, 2528 (2012).
37. Z. Ma, Y. Zhang, C. Cao, J. Yuan, Q. Liu, and J. Wang, *Phys. B Condens. Matter* 406, 4620 (2011).
38. Q. Li, C.W. Kartikowati, S. Horie, T. Ogi, T. Iwaki, and K. Okuyama, *Sci. Rep.* 7, 9894 (2017).
39. N. Chujo, F. Kino, K. Kume, T. Aoyama, and M. Fukuda, *J. Jpn. Soc. Powder Powder Metall.* 63, 624 (2016).
40. M. Anhalt, *J. Magn. Magn. Mater.* 320, e366 (2008).
41. P.-W. Jang, B.-H. Lee, and G.-B. Choi, *J. Korean Magn. Soc.* 20, 13 (2010).
42. M. Anhalt, B. Weidenfeller, and J.L. Mattei, *J. Magn. Magn. Mater.* 320, e844 (2008).
43. A. Maruo and H. Igarashi, *IEEE Trans. Magn.* 55, 1 (2019).
44. Y. Guo, J.G. Zhu, J. Zhong, H. Lu, and J.X. Jin, *IEEE Trans. Magn.* 44, 279 (2008).